

Preface

These are notes for a course on the representation theory of the symmetric group. This topic combines combinatorics, abstract algebra, and linear algebra to understand how representations of the symmetric group break into irreducible components. Second references include. . .

Contents

Contents	i
1 Representation of Finite Groups	1
1.1 Matrix Representations	1
1.1.1 Basic Definitions & Examples	1
1.1.2 New Representations from Old	4
1.1.3 Irreducible Representations	7
1.2 Characters	7
1.2.1 Basic Definitions & Examples	7
1.2.2 Conjugacy Classes & Integer Partitions	9
1.2.3 Class Functions & Inner Products	10
Exercises	12

Chapter 1

Representation of Finite Groups

§ 1.1. Matrix Representations

I. Basic Definitions & Examples

Definition 1.1.1. Let G be a finite group. Let $GL_n(\mathbb{C})$ be the set of $n \times n$ invertible matrices over $\mathbb{C}$. A *degree n (matrix) representation* of G is a homomorphism $X : G \rightarrow GL_n(\mathbb{C})$.

Example 1.1.1. Let G be any group. Let $\mathbb{1} : G \rightarrow GL_1(\mathbb{C})$ be defined by $\mathbb{1}(g) = 1$ for all $g \in G$. This is the *trivial representation*.

Example 1.1.2. Let $X : \mathbb{Z}_2 \times \mathbb{Z}_2 \rightarrow GL_2(\mathbb{C})$ be defined by:

$$\begin{aligned} X((0,0)) &:= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ X((1,0)) &:= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ X((0,1)) &:= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \end{aligned}$$

Note that $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ are their own inverses, as are $(1,0), (0,1) \in \mathbb{Z}_2 \times \mathbb{Z}_2$. The last element is predetermined as:

$$X((1,1)) := \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Recall 1.1.1. The *sign* of a permutation is a function $\text{sign} : S_n \rightarrow \{\pm 1\}$ defined by $\text{sign}(\sigma) := (-1)^m$, where m is the number of transpositions in the permutations decomposition. For example, $(2\ 3\ 6\ 4) = (2\ 3)(3\ 6)(6\ 4)$, hence $\text{sign}((2\ 3\ 6\ 4)) = (-1)^3 = -1$.

Example 1.1.3. Let $X : S_n \rightarrow GL_1(\mathbb{C})$ be defined by $X(\sigma) := \text{sign}(\sigma)$. This is a representation, as the sign function is multiplicative. Indeed, if σ is a k -transposition and τ is a t -transposition, then $\sigma\tau$ is a $k+t$ transposition. This proves that $\text{sign}(\sigma)\text{sign}(\tau) = \text{sign}(\sigma\tau)$, as even + even = even, even + odd = odd, odd + even = odd, and odd + odd = even.

Recall 1.1.2. A *group action* of a group G on a set A is a map $G \times A \rightarrow A$ (written $g.a$ for all $g \in G$ and $a \in A$) satisfying the following properties:

- $g_1.(g_2.a) = (g_1g_2).a$ for all $g_1, g_2 \in G$, $a \in A$, and
- $1.a = a$ for all $a \in A$.

Example 1.1.4. Let $S = \{\{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}\}$ and $G = S_4$. A group action of S_4 on S can be defined by the group element of S_4 permuting a two-element set of S . For example:

$$\begin{aligned}(12).\{1, 3\} &:= \{2, 3\}, \\ (1234).\{1, 2\} &:= \{2, 3\}.\end{aligned}$$

This is indeed a group action, as:

$$\begin{aligned}(12).((1234).\{1, 2\}) &= (12).\{2, 3\} = \{1, 3\} \\ ((12)(1234)).\{1, 2\} &= (234).\{1, 2\} = \{1, 3\}.\end{aligned}$$

Example 1.1.5. Given S from the previous example, we can define an action on S by $\mathbb{Z}_4$. First, note that if σ is a 4-cycle, then $\mathbb{Z}_4 \rightarrow \langle \sigma \rangle$ is an isomorphism given by $[k] \mapsto \sigma^k$. This means if we fix a 4-cycle, we can define an action on S by $\mathbb{Z}_4$. For example, we can define:

$$\begin{aligned}[0].\{1, 2\} &:= (1).\{1, 2\} = \{1, 2\} \\ [1].\{1, 2\} &:= (1234).\{1, 2\} = \{2, 3\} \\ [2].\{1, 2\} &:= (1234)^2.\{1, 2\} = (1234).\{2, 3\} = \{3, 4\} \\ [3].\{1, 2\} &:= (1234)^3.\{1, 2\} = \{1, 4\}.\end{aligned}$$

Example 1.1.6. Let:

$$V := \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : x_1 + x_2 + x_3 = 0 \right\}$$

be a subspace of $\mathbb{C}^3$. Given the set information, we can write $x_1 = -x_2 - x_3$, hence there are two-free variables.

If $x_2 = 1$ and $x_3 = 0$, then one of our basis elements is $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$. If $x_2 = 0$ and $x_3 = 1$, then our other basis element

is $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$.

The group S_3 acts on V by permuting coordinates. For example:

$$(123) \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_3 \\ x_1 \end{bmatrix}.$$

Since V is a two-dimensional subspace of $\mathbb{C}$ with basis $\left\{ v_1 := \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, v_2 := \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$, we can construct a representation $X : V \rightarrow GL_2(\mathbb{C})$ as follows. First, note that:

$$\begin{aligned}(1).v_1 &= v_1 = (1)v_1 + (0)v_2, \\ (1).v_2 &= v_2 = (0)v_1 + (1)v_2.\end{aligned}$$

Hence define:

$$X((1)) := \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Similarly,

$$\begin{aligned} (1\ 2).v_1 &= (1\ 2) \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = -v_1 = (-1)v_1 + (0)v_2, \\ (1\ 2).v_2 &= (1\ 2) \cdot \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = -v_1 + v_2 = (-1)v_1 + (1)v_2. \end{aligned}$$

Thus:

$$X((1\ 2)) := \begin{bmatrix} -1 & 0 \\ -1 & 1 \end{bmatrix}.$$

This can be done for all 9 elements of S_3 . One must then verify by 36 different computations to show that X is indeed a group homomorphism. This is tedious, and instead we will show that a group action *induces* a group homomorphism.

Theorem 1.1.1. *Let G act on $S := \{s_1, \dots, s_n\}$. Define $X : G \rightarrow GL_n(\mathbb{C})$ by:*

$$X(g)_{ij} := \begin{cases} 1, & g.s_j = s_i \\ 0, & \text{otherwise} \end{cases}.$$

Then X is a degree n representation.

Proof. Let V be the (free) vector space over $\mathbb{C}$ with basis S . Note that linear maps are defined by their action on a basis. Fix an element $g \in G$. We define a linear map that sends s_j to $g.s_j$ for all $j = 1, \dots, n$. Then the matrix for this linear map is X . Since $e.s_j = s_j$ and $g.(h.s_j) = (gh).s_j$, then $X(e) = I_n$ and $X(gh) = X(g)X(h)$. Thus X is a homomorphism. ■

Example 1.1.7. Consider the set S from Example ???. Let $s_1 := \{1, 2\}$, $s_2 := \{1, 3\}$, and so on. Given the representation defined in Theorem ??, if we want to compute $X((4\ 3\ 2\ 1))$ then consider:

$$\begin{aligned} (4\ 3\ 2\ 1).s_1 &= (4\ 3\ 2\ 1).\{1, 2\} = \{1, 4\} = s_3, \\ (4\ 3\ 2\ 1).s_2 &= (4\ 3\ 2\ 1).\{1, 3\} = \{2, 4\} = s_5, \\ (4\ 3\ 2\ 1).s_3 &= (4\ 3\ 2\ 1).\{1, 4\} = \{3, 4\} = s_6, \\ &\vdots \end{aligned}$$

Since $(4\ 3\ 2\ 1).s_1 = s_3$, there will be a 1 in the third row and first column of $X((4\ 3\ 2\ 1))_{ij}$. Hence:

$$X((4\ 3\ 2\ 1)) = \begin{matrix} & \begin{matrix} s_1 & s_2 & s_3 & s_4 & s_5 & s_6 \end{matrix} \\ \begin{matrix} s_1 \\ s_2 \\ s_3 \\ s_4 \\ s_5 \\ s_6 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}.$$

Definition 1.1.2. The *defining representation* of S_n , denoted by D , is the representation induced by the action of S_n on $\{1, \dots, n\}$.

Example 1.1.8. Consider the element $(1\ 3\ 2) \in S_4$. We have:

$$\begin{aligned}(1\ 3\ 2).1 &= 3, \\ (1\ 3\ 2).2 &= 1, \\ (1\ 3\ 2).3 &= 2, \\ (1\ 3\ 2).4 &= 4.\end{aligned}$$

Hence:

$$\chi((1\ 3\ 2)) = \begin{matrix} & 1 & 2 & 3 & 4 \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}.$$

The defining representation can also be interpreted as rearranging the rows of an identity matrix. Since the element we are considering is $(1\ 3\ 2)$, we can see that the first row was sent to the third, the third row was sent to the second, and the second row was sent to the first. The fourth row of the identity matrix was not changed.

Definition 1.1.3. The *left regular representation* L of a group $G = \{g_1, \dots, g_n\}$ is the representation induced by the group action on G via left multiplication.

Example 1.1.9. Consider the element $(1, 0) \in \mathbb{Z}_2 \times \mathbb{Z}_2$. We have:

$$\begin{aligned}(1, 0) + (0, 0) &= (1, 0), \\ (1, 0) + (1, 0) &= (0, 0), \\ (1, 0) + (0, 1) &= (1, 1), \\ (1, 0) + (1, 1) &= (0, 1).\end{aligned}$$

Hence:

$$L((1, 0)) = \begin{matrix} & \begin{matrix} (0,0) & (1,0) & (0,1) & (1,1) \end{matrix} \\ \begin{matrix} (0,0) \\ (1,0) \\ (0,1) \\ (1,1) \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Theorem 1.1.2. Let L be the left regular representation of $G = \{g_1, \dots, g_n\}$. Then the matrices $L(g_1), \dots, L(g_n)$ are linearly independent.

Proof. ■

II. New Representations from Old

Proposition 1.1.3. Let $X : G \rightarrow GL_n(\mathbb{C})$ be a representation. Then $\bar{X} : G \rightarrow GL_n(\mathbb{C})$ given by $\bar{X}(g) := \overline{X(g)}$ is a representation.

Proof. ■

Proposition 1.1.4. Let $X : G \rightarrow GL_n(\mathbb{C})$ be a representation. Let $T \in GL_n(\mathbb{C})$. Then $TX(g)T^{-1}$ is a representation.

Proof. ■

Proposition 1.1.5. Let $X : G \rightarrow GL_n(\mathbb{C})$ and $Y : G \rightarrow GL_m(\mathbb{C})$ be matrix representations. Then $X \oplus Y : G \rightarrow GL_{n+m}(\mathbb{C})$ defined by:

$$(X \oplus Y)(g) := \begin{bmatrix} X(g) & 0 \\ 0 & Y(g) \end{bmatrix}$$

is a matrix representation.

Proof. ■

Definition 1.1.4. Given $A = (a_{ij})$ and $B = (b_{ij})$, the *tensor product* $A \otimes B$ is defined as the block matrix whose ij^{th} block is $a_{ij}B$.

Example 1.1.10. Let:

$$A = \begin{bmatrix} 2 & -1 \\ 4 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}.$$

Then:

$$A \otimes B = \left[\begin{array}{ccc|ccc} 2 & 4 & 6 & -1 & -2 & -3 \\ 8 & 10 & 12 & -4 & -5 & -6 \\ 14 & 16 & 18 & -7 & -8 & -9 \\ \hline 4 & 8 & 12 & 0 & 0 & 0 \\ 16 & 20 & 24 & 0 & 0 & 0 \\ 28 & 32 & 36 & 0 & 0 & 0 \end{array} \right].$$

Theorem 1.1.6. If A and B are matrices, then $(AB) \otimes (CD) = (A \otimes C)(B \otimes D)$.

Proof. ■

Proposition 1.1.7. Let $X : G \rightarrow GL_m(\mathbb{C})$ and $Y : H \rightarrow GL_n(\mathbb{C})$ be matrix representations. Then $(X \otimes Y) : G \times H \rightarrow GL_{mn}(\mathbb{C})$ defined by $(X \otimes Y)((g, h)) := X(g) \otimes Y(h)$ is a representation. If $G = H$, then $(X \otimes Y) : G \rightarrow GL_{mn}(\mathbb{C})$ defined by $(X \otimes Y)(g) := X(g) \otimes Y(g)$ is also a representation.

Proof. ■

Example 1.1.11. Let D be the defining representation for S_3 , $\mathbb{1}$ the trivial representation of S_3 , and $Y : S_3 \rightarrow GL_1(\mathbb{C})$ be given by $Y(\sigma) := \text{sign}(\sigma)$. For $(1\ 3\ 2) \in S_3$, we have:

$$\begin{aligned} D((1\ 3\ 2)) &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \\ \mathbb{1}((1\ 3\ 2)) &= [1] \\ Y((1\ 3\ 2)) &= [-1]. \end{aligned}$$

Then:

$$\begin{aligned}
 (D \otimes (1 \oplus Y))((132)) &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
 &= \left[\begin{array}{cc|cc|cc} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ \hline 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{array} \right].
 \end{aligned}$$

Proposition 1.1.8. Let H be a subgroup of G and $X : G \rightarrow GL_n(\mathbb{C})$ a representation. Then $(X \downarrow_H^G) : H \rightarrow GL_n(\mathbb{C})$ is a representation.

Proof. ■

Example 1.1.12. Consider the subgroup $H := \{(1), (123), (132)\}$ of S_3 . Let L be the left regular representation for S_3 . Given $(123) \in H$ we have:

$$\begin{aligned}
 (L \downarrow_H^{S_3} (123)) &= (L(123)) \\
 &= \begin{matrix} & \begin{matrix} (1) & (123) & (132) & (12) & (13) & (23) \end{matrix} \\ \begin{matrix} (1) \\ (123) \\ (132) \\ (12) \\ (13) \\ (23) \end{matrix} & \left[\begin{array}{ccc|ccc} 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array} \right] \end{matrix}.
 \end{aligned}$$

Proposition 1.1.9. Let H be a subgroup of G and let $g_1H, \dots, g_kH$ be the left cosets of H . Let $X : H \rightarrow GL_n(\mathbb{C})$ be a representation. Then $(X \uparrow_H^G) : G \rightarrow GL_{nk}(\mathbb{C})$ defined by:

$$(X \uparrow_H^G(g))_{ij} := \begin{cases} X(g_i^{-1}gg_j), & gg_j \in g_iH \\ 0, & gg_j \notin g_iH \end{cases}$$

is a representation, where ij is a $k \times k$ block given by $X(g_i^{-1}gg_j)$ or 0.

Proof. See Exercise ?? ■

Example 1.1.13. Let D be the defining representation of S_4 . Let $\epsilon A_4, (12)A_4$ be the left cosets of A_4 in S_4 . Given $(1234) \in S_4$, we have:

$$(D \downarrow_{A_4}^{S_4}) \uparrow_{A_4}^{S_4} (1234) = \begin{matrix} & \begin{matrix} \epsilon A_4 & (12)A_4 \end{matrix} \\ \begin{matrix} \epsilon A_4 \\ (12)A_4 \end{matrix} & \left[\begin{array}{c|c} D \downarrow_{A_4}^{S_4} (\epsilon^{-1}(1234)\epsilon) & D \downarrow_{A_4}^{S_4} (\epsilon^{-1}(1234)(12)) \\ \hline D \downarrow_{A_4}^{S_4} ((12)^{-1}(1234)\epsilon) & D \downarrow_{A_4}^{S_4} ((12)^{-1}(1234)(12)) \end{array} \right] \end{matrix}$$

Since $\epsilon^{-1}(1234)\epsilon \notin A_4$, it must be that $D \downarrow_{A_4}^{S_4} (\epsilon^{-1}(1234)\epsilon)$ is the zero block. Similarly, since $(12)^{-1}(1234)(12) \notin A_4$, it must be that $D \downarrow_{A_4}^{S_4} ((12)^{-1}(1234)(12))$ is the zero block. We have $\epsilon^{-1}(1234)(12) = (134) \in A_4$ and

$(12)^{-1}(1234)\epsilon = (234) \in A_4$. Thus:

$$(D \downarrow_{A_4}^{S_4}) \uparrow_{A_4}^{S_4} (1234) = \left[\begin{array}{cccc|cccc} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ \hline 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

Example 1.1.14. Let $G = \{g_1, \dots, g_n\}$ be a group, let L be the left regular representation of G , and let $\mathbb{1}$ denote the trivial representation. Then $L = \mathbb{1} \uparrow_{\{\epsilon\}}^G$. Indeed, the cosets of $\{\epsilon\}$ in G are $g_1, g_2, \dots, g_n$. For any $g \in G$, note that each ij block of $\mathbb{1} \uparrow_{\{\epsilon\}}^G(g)$ is 1×1 . In particular:

$$\begin{aligned} (\mathbb{1} \uparrow_{\{\epsilon\}}^G(g))_{ij} &= \begin{cases} \mathbb{1}(g_i^{-1}g_j), & gg_j \in g_i\{\epsilon\} \\ 0, & gg_j \notin g_i\{\epsilon\} \end{cases} \\ &= \begin{cases} \mathbb{1}(g_i^{-1}g_j), & gg_j = g_i \\ 0, & gg_j \neq g_i \end{cases} \\ &= (L(g))_{ij}. \end{aligned}$$

Example 1.1.15. If G is a group and X is a representation of G , then $X \uparrow_G^G = X$. **prove later**

III. Irreducible Representations

Definition 1.1.5. Let $X : G \rightarrow GL_n(\mathbb{C})$ be a representation. Then X is *reducible* if there exists representations A and B such that X is similar to $A \oplus B$.

§ 1.2. Characters

I. Basic Definitions & Examples

Recall 1.2.1.

- If $(a_{ij}) \in \text{Mat}_n(\mathbb{C})$, the *trace* of A is $\text{tr}(A) = \sum_{i=1}^n a_{ii}$.
- $\text{tr}(AB) = \text{tr}(BA)$
- If $A, B \in \text{Mat}_n(\mathbb{C})$ are similar, then $\text{tr}(A) = \text{tr}(B)$.
- If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of $A \in \text{Mat}_n(\mathbb{C})$, then $\text{tr}(A) = \sum_{i=1}^n \lambda_i$.

Definition 1.2.1. Let $X : G \rightarrow GL_n(\mathbb{C})$ be a representation. The *character* of X is the function $\chi^X : G \rightarrow \mathbb{C}$ defined by $\chi^X(g) := \text{tr}(X(g))$.

Theorem 1.2.1. If representations X and Y are similar, then $\chi^X = \chi^Y$.

Proof. For any $g \in G$:

$$\begin{aligned}
 \chi^X(g) &= \text{tr}(X(g)) \\
 &= \text{tr}(T^{-1}Y(g)T) \\
 &= \text{tr}(T^{-1}(Y(g)T)) \\
 &= \text{tr}(Y(g)TT^{-1}) \\
 &= \text{tr}(Y(g)) \\
 &= \chi^Y(g). \quad \blacksquare
 \end{aligned}$$

Example 1.2.1. Let $G = \mathbb{Z}_4$ and L the left regular representation. Note that:

$$\begin{aligned}
 L(0) &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 L(1) &= \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}
 \end{aligned}$$

Hence $\chi^L(1) = 4$ and $\chi^L(1) = 0$. Note that $\chi^L(2) = \chi^L(3) = 0$, as there does not exist a nonzero element $x \in \mathbb{Z}_4$ with $1 + x = x$. In general, $g \cdot g' \neq g'$ unless $g = e$!

Proposition 1.2.2. Let X , A , and B be representations.

- (1) $\chi^{T^X T^{-1}} = \chi^X$.
- (2) $\chi^{A \oplus B} = \chi^A + \chi^B$.
- (3) $\chi^{A \otimes B} = \chi^A \chi^B$.
- (4) $\chi^{X \uparrow_H^G}(g_0) = \frac{1}{|H|} \sum_{g \in G} \chi^X(g^{-1} g_0 g)$.

Proof. (4) Recall that $X \uparrow_H^G(g_0)$ is a block matrix with entries $X(g_i^{-1} g_0 g_i)$ if $g_i^{-1} g_0 g_i \in H$ and 0 otherwise. Hence:

$$\begin{aligned}
 \chi^{X \uparrow_H^G}(g_0) &= \text{Tr}(X \uparrow_H^G(g_0)) \\
 &= \sum_{i=1}^k \text{Tr}(X(g_i^{-1} g_0 g_i)).
 \end{aligned}$$

Note that this sum is independent of $h \in H$. Hence:

$$\begin{aligned}
 \chi^{X \uparrow_H^G}(g_0) &= \sum_{i=1}^k \text{Tr}(X(g_i^{-1} g_0 g_i)) \\
 &= \frac{1}{|H|} \sum_{h \in H} \sum_{i=1}^k \text{Tr}(X(g_i^{-1} g_0 g_i)).
 \end{aligned}$$

Since the trace is independent of matrix similarity, we can write:

$$\begin{aligned}
 \chi^{X \uparrow_H^G}(g_0) &= \frac{1}{|H|} \sum_{h \in H} \sum_{i=1}^k \text{Tr}(X(g_i^{-1} g_0 g_i)) \\
 &= \frac{1}{|H|} \sum_{h \in H} \sum_{i=1}^k \text{Tr}(X(h^{-1} g_i^{-1} g_0 g_i h)).
 \end{aligned}$$

Since hg_i gives every group element in G , reindexing yields:

$$\begin{aligned}\chi^{X \uparrow_H^G}(g_0) &= \frac{1}{|H|} \sum_{h \in H} \sum_{i=1}^k \text{Tr}(X(h^{-1}g_i^{-1}g_0g_ih)) \\ &= \frac{1}{|H|} \sum_{h' \in G} \text{Tr}(X((h')^{-1}g_0h')).\end{aligned}$$

■

Example 1.2.2. Consider $A_4 \leq S_4$. Let D be the defining representation of S_4 and let $X = D \downarrow_{A_4}^{S_4}$. Then:

$$\begin{aligned}\chi^{X \uparrow_{A_4}^{S_4}}(12) &= \frac{1}{|A_4|} \sum_{\sigma \in A_4} \chi^X(\sigma^{-1}(12)\sigma) \\ &= \frac{1}{|A_4|} \sum_{\sigma \in S_4} \text{Tr}(X(\sigma^{-1}(12)\sigma)).\end{aligned}$$

Note that cycle type is preserved via conjugation. Hence for all $\sigma \in S_4$, it must be that σ^{-1} is a 2-cycle. A single 2-cycle is an odd permutation, hence $\sigma^{-1}(12)\sigma \notin A_4$ for any $\sigma \in S_4$. Thus $\chi^{X \uparrow_{A_4}^{S_4}}(12) = 0$. Now consider:

$$\chi^{X \uparrow_{A_4}^{S_4}}(123) = \frac{1}{|A_4|} \sum_{\sigma \in S_4} \text{Tr}(X(\sigma^{-1}(123)\sigma)).$$

Again, since cycle type is preserved under conjugation, $\sigma^{-1}(123)\sigma$ will be a 3-cycle. Note that the defining representation of a permutation σ permutes the rows of the identity matrix according to σ . Hence $\sigma^{-1}(123)\sigma$ will leave exactly one row fixed, meaning that $\text{Tr}(X(\sigma^{-1}(123)\sigma)) = 1$ for all $\sigma \in S_4$. Thus $\chi^{X \uparrow_{A_4}^{S_4}}(123) = \frac{24}{|A_4|} = 2$.

II. Conjugacy Classes & Integer Partitions

Definition 1.2.2. If G is a group, the *conjugacy class* of $g \in G$ is $C_g := \{h^{-1}gh : h \in G\}$.

Example 1.2.3. The conjugacy class of $C_{(123)} \subset S_4$ is:

$$\begin{aligned}C_{(123)} &= \{\sigma^{-1}(123)\sigma : \sigma \in S_4\} \\ &= \{\text{all 3-cycles in } S_4\}.\end{aligned}$$

Recall 1.2.2. Conjugacy in A_4 is weird. Conjugacy just swaps the order of the middle term. For example:

$$(12)^{-1}(123)(12) = (212).$$

Example 1.2.4. If G is abelian, then $C_g = \{g\}$.

Theorem 1.2.3. Characters are constant on conjugacy classes.

Proof. If X is a representation, then:

$$\begin{aligned}\chi^X(h^{-1}gh) &= \text{Tr}(X(h^{-1}gh)) \\ &= \text{Tr}(X(h)^{-1}X(g)X(h)) \\ &= \text{Tr}(X(g)) \\ &= \chi^X(g).\end{aligned}$$

■

Definition 1.2.3. An *integer partition* of n , denoted $\sigma \vdash n$, is a weakly decreasing list of non-negative integers that sum to n . We define $\max \lambda$ to be the max list element and $\ell(\lambda)$ the number of nonzero list elements.

Example 1.2.5. For $\lambda \vdash 5$, we have:

$\lambda \vdash 5$	$\max \lambda$	$\ell(\lambda)$	$1^{m_1} 2^{m_2} 3^{m_3} \dots$	Young diagram
(5)	5	1	5^1	
(4, 1)	4	2	$1^1 4^1$	
(3, 2)	3	2	$2^1 3^1$	
(3, 1, 1)	3	3	$1^2 3^1$	
(2, 2, 1)	2	3	$1^1 2^2$	
(2, 1, 1, 1)	2	4	$1^3 2^1$	
(1, 1, 1, 1, 1)	1	5	1^5	

Example 1.2.6. We can use integer partitions to describe the cycle types of elements in S_n . For example, the cycle type of $(12)(345)(6)(78) \in S_8$ is $(3, 2, 2, 1)$.

Example 1.2.7. Recall that $S_3 = \{(1), (12), (13), (23), (123), (132)\}$. The conjugacy classes of S_3 are:

$$\begin{aligned} C_{(1)} &= \{(1)\}, \\ C_{(12)} &= \{(12), (13), (23)\}, \\ C_{(123)} &= \{(123), (132)\}. \end{aligned}$$

Definition 1.2.4. Let $\lambda = 1^{m_1} 2^{m_2} \dots \alpha^{m_\alpha} \vdash n$. We define $z_\lambda := 1^{m_1} m_1! \cdot 3^{m_3} m_3! \cdot \dots \cdot \alpha^{m_\alpha} m_\alpha!$.

Theorem 1.2.4. Let $\sigma \in S_n$ have cycle type $\lambda = 1^{m_1} 2^{m_2} \dots \alpha^{m_\alpha} \vdash n$. Then $|C_\sigma| = \frac{n!}{z_\lambda}$.

Proof. There are $n!$ choices of permutations in S_n in one line notation. Place parenthesis around the integers to create a permutation of cycle type λ . By the way λ is defined, there are m_2 different 2-cycles. Note that each 2-cycle can be rewritten in two different ways. Similarly, each 3-cycle can be rewritten in three different ways. Hence we've overcounted by $1^{m_1} 2^{m_2} \dots \alpha^{m_\alpha}$. But note that there are m_1 1-cycles, and each can be permuted throughout all of their positions. Hence we've overcounted by $m_1!$. Similarly, there are m_2 2-cycles, and each can be permuted throughout all of their positions. Hence we've overcounted by $m_2!$. So in total, we've overcounted by $1^{m_1} 2^{m_2} \dots \alpha^{m_\alpha} m_1! m_2! \dots m_\alpha! = z_\lambda$. Thus $|C_\sigma| = \frac{n!}{z_\lambda}$. ■

III. Class Functions & Inner Products

Definition 1.2.5. Let G be a group. A *class function* is a function $f : G \rightarrow \mathbb{C}$ which is constant on the conjugacy classes of G . The set of all class functions is denoted $C(G)$.

Example 1.2.8. Class functions can be trivially defined. For instance, define $f : S_3 \rightarrow \mathbb{C}$ by:

$$\begin{aligned} f((1)) &= 1 \\ f((12)) &= f((13)) = f((23)) = 2 \\ f((123)) &= f((132)) = 3. \end{aligned}$$

Then $f \in C(S_3)$.

Proposition 1.2.5. If $C_{g_1}, \dots, C_{g_n}$ are the conjugacy classes of G , then $\{\mathbb{1}_{C_{g_1}}, \dots, \mathbb{1}_{C_{g_n}}\}$ is a basis for $C(G)$. In particular, the dimension of $C(G)$ is the number of conjugacy classes of G .

Proof.

■

Example 1.2.9. A basis for $C(S_3)$ is $\{1_\lambda : \lambda \vdash 3\}$, where:

$$1_\lambda(\sigma) = \begin{cases} 1, & \text{if } \sigma \text{ is the same cycle type as } \lambda \\ 0, & \text{otherwise} \end{cases}.$$

Theorem 1.2.6. $C(G)$ is an inner product space, with inner product defined by:

$$\langle f, g \rangle_G := \frac{1}{|G|} \sum_{h \in G} f(h) \overline{g(h)}.$$

Proof. ■

Example 1.2.10. Let L be the left regular representation and D the defining representation of S_4 . Then:

$$\langle \chi^L, \chi^D \rangle_{S_4} = \frac{1}{|S_4|} \sum_{\sigma \in S_4} \chi^L(\sigma) \overline{\chi^D(\sigma)}.$$

Note that $\text{im } \chi^D \subset \mathbb{R}$, hence $\overline{\chi^D(\sigma)} = \chi^D(\sigma)$ for all $\sigma \in S_4$. With this:

$$\begin{aligned} \langle \chi^L, \chi^D \rangle_{S_4} &= \frac{1}{|S_4|} \sum_{\sigma \in S_4} \chi^L(\sigma) \chi^D(\sigma) \\ &= \frac{1}{|S_4|} \sum_{\sigma \in S_4} \text{Tr}(L(\sigma)) \text{Tr}(D(\sigma)). \end{aligned}$$

Note that $L(\epsilon) = I_{24}$ and $D(\epsilon) = I_4$. Hence $\text{Tr}(L(\epsilon)) \text{Tr}(D(\epsilon)) = 96$. Moreover, for any $\sigma \neq \epsilon$, there does not exist an element τ with $\sigma\tau = \tau$. Hence, for all $\sigma \neq \epsilon$, $\text{Tr}(L(\sigma)) = 0$. Thus:

$$\begin{aligned} \langle \chi^L, \chi^D \rangle_{S_4} &= \frac{1}{24} \cdot 96 \\ &= 4. \end{aligned}$$

Exercises

Exercise 1.

- (a) Let E be an $n \times n$ matrix over $\mathbb{C}$ such that $E \neq I_n$, $E \neq 0$, and $E^2 = E$. Show that there is an $n \times n$ matrix T such that:

$$E = T^{-1} \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} T$$

for some k .

- (b) Let G be a group and X be a function mapping G into the set of (possible singular) $n \times n$ matrices over $\mathbb{C}$ such that $X(e) \neq 0$ and $X(gh) = X(g)X(h)$ for all $g, h \in G$. Show that there is an $n \times n$ matrix T and a matrix representation Y of G such that:

$$X(g) = T^{-1} \begin{bmatrix} Y(g) & 0 \\ 0 & 0 \end{bmatrix} T$$

for all $g \in G$.

Proof. (a) Since E is a projection, its eigenvalues are $\lambda = 0$ and $\lambda = 1$. Let $E_0 := \ker(E)$ and $E_1 := \ker(E - I_n)$ denote the eigenspaces of $\lambda = 0$ and $\lambda = 1$. Note that $Ev \in E_1$, as:

$$\begin{aligned} (E - I_n)(Ev) &= E^2v - Ev \\ &= 0. \end{aligned}$$

Similarly, $v - Ev \in E_0$ since:

$$\begin{aligned} E(v - Ev) &= Ev - E^2v \\ &= 0. \end{aligned}$$

Since every $v \in V$ can be written as $(v - Ev) + Ev$, this means $V = E_0 + E_1$. If $x \in E_0 \cap E_1$, then $Ex = 0$ and $Ex = x$. It must be the case that $x = 0$, hence $V = E_0 \oplus E_1$. Since V is the direct sum of eigenspaces, E is diagonalizable. Assuming that $\dim(E_1) = k$ and $\dim(E_0) = n - k$, there exists a matrix T such that

$$E = T^{-1} \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} T.$$

- (b) Note that $X(e)^2 = X(e)X(e) = X(e)$. Hence by part (a) there exists an $n \times n$ matrix T such that:

$$X(g) = T^{-1} \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} T.$$

Note that:

$$TX(g)T^{-1} = (TX(g)T^{-1})(TX(e)T^{-1}) = (TX(g)T^{-1}) \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix}.$$

Similarly:

$$TX(g)T^{-1} = (TX(e)T^{-1})(TX(g)T^{-1}) = \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} (TX(g)T^{-1}).$$

Since $(TX(g)T^{-1}) \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} (TX(g)T^{-1})$, it must be the case that $TX(g)T^{-1} = \begin{bmatrix} Y(g) & 0 \\ 0 & 0 \end{bmatrix}$ for some $Y(g) \in \text{Mat}_k(\mathbb{C})$. It remains to show that Y is a representation. Note that:

$$\begin{bmatrix} Y(e) & 0 \\ 0 & 0 \end{bmatrix} = TX(e)T^{-1} = \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix}.$$

Thus $Y(\epsilon) = I_k$. Similarly:

$$\begin{aligned} \begin{bmatrix} Y(gh) & 0 \\ 0 & 0 \end{bmatrix} &= T X(gh) T^{-1} \\ &= (T X(g) T^{-1})(T X(h) T^{-1}) \\ &= \begin{bmatrix} Y(g) & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} Y(h) & 0 \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} Y(g)Y(h) & 0 \\ 0 & 0 \end{bmatrix}. \end{aligned}$$

Thus $Y(gh) = Y(g)Y(h)$. It remains to show that $\text{im } Y = \text{GL}_n(\mathbb{C})$. Indeed, for any $Y(g) \in \text{im } Y$ we have $Y(\epsilon) = Y(gg^{-1}) = Y(g)Y(g^{-1})$, hence $Y(g)$ does have an inverse. Thus, we've shown that there exists an $n \times n$ matrix T and matrix representation Y of G such that:

$$X(g) = T^{-1} \begin{bmatrix} Y(g) & 0 \\ 0 & 0 \end{bmatrix} T \quad \blacksquare$$

Exercise 2. For $\sigma = \sigma_1 \dots \sigma_n \in S_n$, written in one line notation, $\text{inv } \sigma$ is the number of pairs $i < j$ such that $\sigma_i > \sigma_j$. Define $\text{sign } \sigma := (-1)^{\text{inv } \sigma}$. Show that switching the position of any two integers in σ changes $\text{inv } \sigma$ by an odd number. This fact implies:

$$\text{sign}(\sigma\tau) = \text{sign } \sigma \text{sign } \tau,$$

where τ is any transposition. Since any permutation can be written as a product of transpositions, this means that the sign representation $X(\sigma) = \text{sign}(\sigma)$ is indeed a matrix representation of S_n .

Proof. Let τ be obtained from σ by swapping the entries in positions $p < q$. After the swap, suppose $\sigma_p < \sigma_q$. Let $m = \text{card}\{r : p < r < q \text{ and } \sigma_p < \sigma_r < \sigma_q\}$. Any inversions involving neither p nor q is unchanged, hence the total number of inversions is unchanged. Entries outside the interval from p to q also leave the total number of inversions unchanged. The pair (p, q) was not an inversion in σ , but becomes one in τ , contributing $+1$. Now consider $p < r < q$. If $\sigma_p < \sigma_r < \sigma_q$, then after the swap (p, r) and (r, q) are inversions. Thus each r contributes $+2$. If $\sigma_r < \sigma_p$ or $\sigma_r > \sigma_q$, then in either case the pairs contribute one inversion both before and after the swap, so there is no change. Thus:

$$\text{inv}(\tau) - \text{inv}(\sigma) = 1 + 2m,$$

which is odd. If $\sigma_p > \sigma_q$, the same argument gives $-(1 + 2m)$ which is also odd.

If $\sigma \in S_n$ and τ is a transposition, then: **DNF** ■

Exercise 3.

Exercise 4. Let $X : H \rightarrow \text{GL}_n(\mathbb{C})$ be a representation and H a subgroup of G . Show that the induced representation $X \uparrow_H^G$ is indeed a representation. asf

Proof. Suppose $g_1H, \dots, g_kH$ are the left cosets of H in G . We have:

$$(X \uparrow_H^G(\epsilon))_{ij} = \begin{cases} X(g_i^{-1}g_j), & g_j \in g_iH \\ 0, & g_j \notin g_iH \end{cases}.$$

Since the left cosets of H are disjoint, for all $i \neq j$ we have $g_j \notin g_iH$. Hence:

$$(X \uparrow_H^G(\epsilon))_{ij} = \begin{cases} X(\epsilon), & i = j \\ 0, & i \neq j \end{cases}.$$

Since X is a representation, $X(\epsilon) = I_n$. Thus:

$$X \uparrow_H^G(\epsilon) = I_{nk}.$$

Now for any $x, y \in G$, we have:

$$\begin{aligned} (X \uparrow_H^G(x) X \uparrow_H^G(y))_{ij} &= \sum_{\alpha=1}^k (X \uparrow_H^G(x))_{i\alpha} (X \uparrow_H^G(y))_{\alpha j} \\ &= \sum_{\alpha=1}^k \left(\begin{cases} X(g_i^{-1} x g_\alpha), & x g_\alpha \in g_i H \\ 0, & x g_\alpha \notin g_i H \end{cases} \right) \left(\begin{cases} X(g_\alpha^{-1} y g_j), & y g_j \in g_\alpha H \\ 0, & y g_j \notin g_\alpha H \end{cases} \right) \end{aligned}$$

Since the left cosets of H are disjoint, there is exactly one α for which $y g_j \in g_\alpha H$. For that α , there is exactly one i such that $x g_\alpha \in g_i H$. In this case, if $x g_\alpha \in g_i H$ and $y g_j \in g_\alpha H$, then it follows that $x y g_j \in g_i H$. Hence:

$$\begin{aligned} (X \uparrow_H^G(x) X \uparrow_H^G(y))_{ij} &= \begin{cases} X(g_i^{-1} x g_\alpha) X(g_\alpha^{-1} y g_j), & x y g_j \in g_i H \\ 0, & x y g_j \notin g_i H \end{cases} \\ &= \begin{cases} X(g_i^{-1} x y g_j), & x y g_j \in g_i H \\ 0, & x y g_j \notin g_i H \end{cases} \\ &= (X \uparrow_H^G(xy))_{ij}. \end{aligned}$$

Thus $X \uparrow_H^G$ is a representation. ■

Exercise 5.

Exercise 6.

Exercise 7. This exercise shows that many of our results do not hold over arbitrary fields. Let X be the representation of $\mathbb{Z}_3 = \{0, 1, 2\}$ generated by:

$$X(1) = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}.$$

Show that X is reducible over $\mathbb{C}$ but not over $\mathbb{R}$.

Proof. Consider $X : \mathbb{Z}_3 \rightarrow GL_2(\mathbb{C})$ as above. Recall that a subspace $V \subseteq \mathbb{C}^2$ is X -invariant if $X(g)v \in V$ for all $v \in V$ and all $g \in \mathbb{Z}_3$. Since $\lambda = \frac{-1 \pm \sqrt{3}}{2}$ are eigenvalues of $X(1)$ and $X(2)$, E_λ is an invariant subspace of $X(1)$ and $X(2)$. Similarly, E_λ is an invariant subspace of $X(0)$ because every subspace of $\mathbb{C}^n$ is invariant under the identity. Thus E_λ is X -invariant, hence X is reducible.

Before considering $X : \mathbb{Z}_3 \rightarrow GL_2(\mathbb{R})$, note that if X is a degree 2 representation, then X is reducible if and only if X is diagonalizable. Indeed, if X were reducible, then X is similar to $A \oplus B$, where A and B are 1×1 block entries of $A \oplus B$. Suppose towards contradiction $X : \mathbb{Z}_3 \rightarrow GL_2(\mathbb{R})$ is reducible. Then X must be diagonalizable. But this is a contradiction, as its characteristic polynomial $x^2 + x + 1$ does not split over $\mathbb{R}$. Hence X is irreducible over $\mathbb{R}$. ■

Exercise 8. This exercise shows that many of our results do not hold for infinite groups.

- (a) Let G be the infinite group $\mathbb{R}^+$ with the operation of multiplication. Show that the function X defined by:

$$X(x) := \begin{bmatrix} 1 & 0 \\ \ln x & 1 \end{bmatrix}$$

for all $x \in \mathbb{R}^+$ is an irreducible representation.

(b) Show that:

$$\text{span} \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

is an X -invariant subspace of $\mathbb{C}^2$, showing Theorem ?? does not hold. Where in the proof of Theorem ?? do we use the finiteness of G ?

Proof. (a) Suppose towards contradiction X is reducible. For any $\alpha \in \mathbb{R}^+$ the characteristic polynomial of $X(\alpha)$ is $(\lambda - 1)^2$. Hence $\lambda = 1$ is an eigenvalue. It follows that $E_1 = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$. Since $\mathbb{C}^2$ doesn't have a basis of eigenvectors, $X(\alpha)$ is not diagonalizable for any $\alpha \in \mathbb{R}^+$. This contradicts the assumption that X is reducible, since if it were reducible it can be written as a diagonal matrix.

(b) For any $\alpha \in \mathbb{R}^+$, we have $X(\alpha) \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Hence $\text{span} \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ is an X -invariant subspace whilst X is irreducible. This shows that Theorem ?? does not hold. In the proof, it may not be the case that $M = \sum_{g \in G} X(g)^* X(g)$ and $v^* M v$ converges, as G is an infinite group. ■

Exercise 9.

Exercise 10.

Exercise 11. Prove that if X and Y are representations of G such that X is degree 1 and Y is irreducible, then $X \otimes Y$ is also an irreducible representation.

Proof. Suppose $Y : G \rightarrow \text{GL}_m(\mathbb{C})$ is an irreducible degree m representation. Note that $(X \otimes Y) : G \rightarrow \text{GL}_m(\mathbb{C})$ is given by $(X \otimes Y)(g) = X(g) \otimes Y(g)$. Since $X(g)$ is degree 1, by definition $X(g) \in \mathbb{C}$, hence $X(g) \otimes Y(g) = X(g)Y(g)$. Suppose towards contradiction $X \otimes Y$ is reducible. Then there exists a subspace $V \subseteq \mathbb{C}^n$ which is $(X \otimes Y)$ -invariant. Hence for all $g \in G$ and $v \in V$, we have $(X \otimes Y)(g)v = X(g)Y(g)v \in V$. But since V is a subspace, it is closed under scalar multiplication. Hence $X(g)^{-1}(X(g)Y(g))v = Y(g)v \in V$. But this means Y is V -invariant, which is a contradiction. Thus $X \otimes Y$ is irreducible. ■